Halvor Hafnor

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Summary

Embedded software and machine learning engineer building classification pipelines for real sensor data on constrained hardware, from collection through feature extraction to on-device inference. Two years at Aerendir Mobile, a shipped automotive product, a pending patent, and an M.S. at UC San Diego.

Experience

Embedded Software and ML Engineer — Jul. 2024 – Present
Aerendir Mobile · part time in term, full time out of term — *Mountain View, CA*

- Evaluation and hardening of the motion-sensor classifiers behind **A-Live**, the anti-bot liveness check, building the lab that measures how model and defense changes perform.
- Built Android applications for real-time biometric data streaming and workflow control.
- Collected and classified the motion dataset behind the company's biometric authentication work: 400 participants, each recorded holding the phone in three postures, sitting, standing and walking, from the accelerometer and gyroscope.
- Prototyped automotive driver monitoring from steering-wheel biometrics, shipped as **AerWheel**. Trained decision-tree classifiers in scikit-learn on features from a 240 Hz six-axis IMU, first through ST's on-sensor Machine Learning Core and then as an external tree running in firmware. Better than **99% accuracy** at under 0.5% false positives, answering within 1.5 seconds. Subject of a pending patent naming me as an inventor.

Electrical Engineering Intern — Jun. 2021 – Sep. 2023
ABB Electrification — *Skien, Norway*

- Assembled and wired around 300 industrial control panels and power cabinets from engineering schematics, across three consecutive summers.
- Supported production and testing of custom electrical systems for industrial clients.

Education

University of California - San Diego - San Diego, CA
Computer Engineering, M.S. — Jan. 2026 – June 2027
Computer Engineering, B.S. Major GPA: 3.9 — Sep. 2024 – June 2026

- **Provost Honors** every quarter enrolled, and **Tau Beta Pi**, the national engineering honor society.

De Anza College - Cupertino, CA — GPA: 4.0
Computer Science & Engineering — Sep. 2022 – June 2024

- Math & Science Tutor and CS Teaching Assistant for calculus, physics, discrete math, C++, Java and Assembly.

Technical Skills

Languages: C++, C, Python, TypeScript, Java, Bash, ARM Assembly
Embedded: Embedded Linux, Android, QML/Qt, STM32, STM32CubeIDE, ST sensor boards, firmware, on-sensor inference, Docker
Signals & ML: PyTorch, scikit-learn, NumPy, pandas, SciPy, OpenCV, YOLOv8, signal processing, feature extraction
Systems: Node, SQLite, WebSocket protocols
Agentic Development: Claude Code, MCP, spec-driven work breakdown, subagent orchestration, verification gates

Projects

TabTerm, a macOS terminal in Chrome tabs

- A Chrome extension over a background terminal daemon that puts every shell session in a real Chrome tab, so tab groups, window management and keyboard shortcuts apply to terminals. Adds a command launcher, history search, project-scoped saved commands and a plugin API.

Laser targeting on an autonomous car, UC San Diego, ECE 148

- Real-time person tracking on a Raspberry Pi 5, in Docker. YOLOv8 detection and OAK-D stereo depth, converted through a 71.9° field of view into pan and tilt angles so two servos hold a laser on the nearest person from 0.5 to 2.9 m. Depth is the median of a 10x10 patch, since one bad pixel throws the aim.